

H-101
Atmospheric crude charge heater
60.7 MW abs. / 67.4 MW fired; 2-cell cabin, 8 passes, 144 rad. tubes 6" x 18.3 m
DESIGN 38.5 barg / 545 C; CA 3 mm
DUTY 60.68 MW
MAT'L Tubes A335 P9 (9Cr-1Mo); convection P5 / CS studded

LP STEAM HEADER
CFU-900-PR-PID-016

FROM E-111 (CIT)
CFU-100-PR-PID-003

BFW HEADER
CFU-900-PR-PID-016
1" BFW-100-072-016-H

TV-1024
FC

TIC
1024

350 C
TT
1024

PI
1227

STRIPPING STEAM HDR
CFU-100-PR-PID-006

TT
1020A-C

2003
SIF-106

TO C-101 FLASH ZONE
CFU-100-PR-PID-006

TT
1020

TIC
1020

SP CASCADE TO PIC-1021
(CFU-100-PR-PID-005); FF FROM FIC-1001

TDIC
1019

BIAS TO FIC-1011-1018
(TOTAL FLOW HELD)

18" P-100-042-B2-H

2003 PER PASS
LOW-LOW PASS FLOW

FT
1011

FIC
1011

FV-1011
FO

FT
1011A-C

FZLL
1011

6" P-100-059-B2-H

FT
1012

FIC
1012

FV-1012
FO

FT
1012A-C

FZLL
1012

6" P-100-060-B2-H

FT
1013

FIC
1013

FV-1013
FO

FT
1013A-C

FZLL
1013

6" P-100-061-B2-H

FT
1014

FIC
1014

FV-1014
FO

FT
1014A-C

FZLL
1014

6" P-100-062-B2-H

FT
1015

FIC
1015

FV-1015
FO

FT
1015A-C

FZLL
1015

6" P-100-063-B2-H

FT
1016

FIC
1016

FV-1016
FO

FT
1016A-C

FZLL
1016

6" P-100-064-B2-H

FT
1017

FIC
1017

FV-1017
FO

FT
1017A-C

FZLL
1017

6" P-100-065-B2-H

FT
1018

FIC
1018

FV-1018
FO

FT
1018A-C

FZLL
1018

6" P-100-066-B2-H

8" P-100-067-B3-H

8" P-100-068-B3-H

8" P-100-069-B3-H

8" P-100-070-B3-H

8" P-100-071-B3-H

8" P-100-072-B3-H

8" P-100-073-B3-H

8" P-100-074-B3-H

8" P-100-075-B3-H

8" P-100-076-B3-H

8" P-100-077-B3-H

8" P-100-078-B3-H

8" P-100-079-B3-H

8" P-100-080-B3-H

8" P-100-081-B3-H

8" P-100-082-B3-H

8" P-100-083-B3-H

8" P-100-084-B3-H

8" P-100-085-B3-H

8" P-100-086-B3-H

8" P-100-087-B3-H

8" P-100-088-B3-H

8" P-100-089-B3-H

CONV.

SS SUPERHEAT COIL

PASS 1

PASS 2

PASS 3

PASS 4

PASS 5

PASS 6

PASS 7

PASS 8

H-101

16 BURNERS (FLOOR-FIRED, 2 CELLS) - SEE CFU-100-PR-PID-005

- NOTES**
- Passes shown schematically (convection + radiant in series per pass); 8 identical passes.
 - Tube-skin thermocouples: 4 per pass (TI-1101..1132) - shown on heater vendor P&ID, alarmed in DCS.
 - Pass flow SIS transmitters FT-10xA/B/C (2003 per pass) independent from BPCS FT-10x (SIF-101).
 - Decoking (steam-air) connections and spool pieces at inlet / outlet manifolds by heater vendor.
 - Line numbers: <NPS>-<fluid>-<area>-<seq>-<class>-<insulation> per CONVENTIONS; see line list CFU-000-PI-LL-001.
 - All vents / drains 3/4" min. with blind or cap; HPV/LPD to piping specification. Valve sizes = line size UNO.
 - Instrument tags per ISA-5.1; index CFU-000-IC-IDX-001. SIS functions per SRS (SIF tags shown in interlocks).

REV	DATE	DESCRIPTION	
A	2026-10-02	Issued for review (FEED)	
GENERIC REFINERY - GULF COAST			
100 KBPSD CRUDE & VACUUM DISTILLATION UNIT			
PIPING & INSTRUMENT DIAGRAM			
ATMOSPHERIC HEATER H-101 - PROCESS COILS			
DISCIPLINE	STAGE	SCALE	SHEET
PROCESS	FEED	NTS	5 OF 17
DRAWN	CHECKED	APPROVED	DATE
CLAUDE CODE	-	-	2026-10-02
DRAWING NUMBER			REV
CFU-100-PR-PID-004			A